

EECE7352 Computer Architecture

Homework 2

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Part E:

Amdahl, Blaauw and Brooks 1964 paper on the IBM 360 Architecture.

The most impressive feature is the introduction of a consistent, upward-compatible architecture across an entire product line (IBM System/360). At the time (1964), computers were often one-off designs, with software tied to a specific machine. The idea that you could write one program and run it on both small and large machines in the family was revolutionary.

Representation of Data Types in system 360

- **Binary Integers:** Supported multiple lengths (8-, 16-, 32-, and 64-bit fixed-point).
- **Decimal Arithmetic:** Included packed decimal support, designed for business applications requiring exact decimal precision (e.g., banking, accounting).
- **Floating-Point:** Provided short (32-bit), long (64-bit), and extended (128-bit) floating-point representations that was very advanced for the era.
- **Character Handling:** Supported 8-bit EBCDIC characters, enabling efficient text processing.

IBM 360 vs RISC -V:

The RISC-V base ISA follows a minimalist design approach. It only supports 32-bit or 64-bit integers (RV32I and RV64I), while advanced features such as floating-point operations, vector processing, and specialized instructions are excluded from the core and instead provided as optional extensions like F and D for floating-point or V for vectors. Unlike System/360, RISC-V offers no native support for decimal arithmetic, leaving such operations to software, and it does not treat character data as a primitive type characters are handled simply as bytes in memory.

Why System 360 stands out:

The System 360 provided a single, unified machine architecture with a rich set of data types that could serve scientific, engineering, and business workloads alike. RISC-V, by contrast, aims to keep the base lean and modular, allowing users to add only the extensions they require, which simplifies implementation and provides flexibility.